

Robot Mules, Birds and Snakes Helping on the Warfront

From drones and other surveillance bots, to robotic surgeons and payload carriers, robots are lending a helping hand to military forces in many countries. But it will be long, or never, before these replace man on the warfront. After all, they have neither heart and mind, nor any reason to fight



LandShark robot from Black-I Robotics

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A recent newspaper headline blared: "Humans lose, robots win in Pentagon's new defence budget." The story was about US president Obama's decision to downsize the army by 100,000 soldiers in a move towards forming a lean but technologically-advanced army. He means to invest in electronic weapons to jam enemy defences and attack online networks, elite commando forces like the ones who rescued two aid workers kidnapped in Somalia, and more than 65 Predator and Reaper combat air patrols.

Predator, Reaper, LandShark, Ar-

madillo, Snakebot... countries all over the world are going in for robot-helpers for their military forces. According to an ABI Research report released last year, around 50 to 80 countries are already using robotic defence systems, or are in the process of building or acquiring the technology to incorporate these into their military programmes.

These robots with the common purpose to replace or supplement humans in battlefields, may take the form of unmanned aerial vehicles (UAVs), unmanned ground vehicles (UGVs) and even unmanned underwater vehicles (UUVs). The report also predicts that the global market for military robotics will grow to more than \$8 billion in 2016.

Soldiers and robots are indeed traversing warfronts side-by-side, and it appears that the trend is strengthening. We just hope they do not come face to face against each other some day, like Asimov's bad dreams.

Marching ahead, unmanned

One of the main reasons for deploying robots on the warfront is to reduce injury and death of soldiers. This advantage obviously can be achieved only by unmanned robots. Several nations' military forces have now started using unmanned vehicles for surveillance, detecting and detonating landmines and bombs, and traversing tough geographic and weather conditions. There are many such aerial, ground and underwater systems present today, but these are still a nascent technology and usually deployed in association with manned vehicles or human control.

"A perfect example of this is shown in the movie *The Hurt Locker*. The movie shows well how such robots are being used in defence. It is safe to say that both manned and unmanned vehicles are used in collaboration in the defence sector," says Praveen Pitchai, who is working towards a masters degree in robotics at the University of Pennsylvania. He is also part of a team working on building a tele-operated/autonomous rover for a NASA competition.

Pitchai explains that there is a lot of work being done with regard to UAVs like the Predator, which is used by many countries for surveillance